

Mechanics

Lecture notes, worked examples and tutorial questions from past Singapore Junior Physics Olympiad papers

LECTURE NOTES · EXAMPLES · TUTORIAL

Name _____

Class _____

Start date _____

Target finish _____

Questions are reproduced from: SJPO 2018 General Round · SJPO 2022 Individual Round · SJPO 2024 · SJPO 2026. Each question title gives its source paper and question number. Questions are cropped directly from the original papers, so their numbering and option labels match the originals.



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How to use this guide. Read each section, then attempt its examples in the dotted working boxes *before* watching the worked-solution video on the e-learning platform. Each subsection has a narrated lecture and a narrated walkthrough of its examples. The tutorial questions at the end have no working space: do them on foolscap paper as timed practice, and check them against the separate Solutions booklet.

1 Kinematics

Describing motion without asking what causes it: vectors, graphs, the equations of uniform acceleration, relative velocity and projectiles.

1.1 Vectors and Measurement

1.3 Relative Motion

1.2 Motion in a Straight Line

1.4 Projectile Motion

LEARNING OUTCOMES

Candidates should be able to:

- add, subtract and resolve vectors, and use the dot and cross products of unit vectors
- state how uncertainties in measurements decide whether two readings can be the same quantity
- interpret displacement–time and velocity–time graphs using gradients and areas
- derive and apply the equations of motion for constant acceleration, including reaction-time problems
- solve relative-velocity problems by changing reference frame
- analyse projectile motion by treating horizontal and vertical motion independently

1.1 Vectors and Measurement

A **scalar** has magnitude only (mass, energy, speed). A **vector** has magnitude and direction (displacement, velocity, force). In Cartesian form we write $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$ with magnitude $|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$.

A **unit vector** $\hat{A} = \vec{A}/|\vec{A}|$ has magnitude exactly 1. Note that $\hat{i} + \hat{j}$ is *not* a unit vector: its magnitude is $\sqrt{2}$.

Dot and cross products

KEY RESULTS

$$\vec{A} \cdot \vec{B} = AB \cos \theta \quad (\text{a scalar}), \quad |\vec{A} \times \vec{B}| = AB \sin \theta \quad (\text{a vector, right-hand rule})$$
$$\hat{i} \cdot \hat{i} = 1, \quad \hat{i} \cdot \hat{j} = 0, \quad \hat{i} \times \hat{j} = \hat{k}, \quad \hat{j} \times \hat{k} = \hat{i}, \quad \hat{k} \times \hat{i} = \hat{j}, \quad \hat{i} \times \hat{i} = \vec{0}$$

Reversing the order of a cross product flips its sign: $\hat{j} \times \hat{i} = -\hat{k}$. Cross products appear throughout mechanics: torque $\vec{\tau} = \vec{r} \times \vec{F}$ and angular momentum $\vec{L} = \vec{r} \times \vec{p}$ (Chapter 8).

Question 1

Which of the following is a unit vector? ($\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$ are standard unit vectors in the x, y, z directions respectively.)

- (a) $\hat{\mathbf{i}} + \hat{\mathbf{j}}$
- (b) $\hat{\mathbf{i}} \cdot \hat{\mathbf{i}}$
- (c) $\hat{\mathbf{i}} \times \hat{\mathbf{i}}$
- (d) $(\hat{\mathbf{i}} \times \hat{\mathbf{j}}) \times \hat{\mathbf{k}}$
- (e) $(\hat{\mathbf{i}} \times \hat{\mathbf{j}}) \times \hat{\mathbf{i}}$

WORKING**Answer** _____**Uncertainty**

A reading $x \pm \Delta x$ means the true value lies somewhere in $[x - \Delta x, x + \Delta x]$. Two readings *could* describe the same object if their ranges overlap. For quantities combined by multiplication or division, fractional uncertainties add: if $Q = A^a B^b$ then $\frac{\Delta Q}{Q} = |a| \frac{\Delta A}{A} + |b| \frac{\Delta B}{B}$.

EXAMPLE 1.2 **SJPO 2018 Q50**

50. Suppose that you own an unmarked gold bar. When you measured its mass, the scale that you used gives a reading of 0.998 kg. The uncertainty of the scale is given as ± 0.005 kg. However, the very next day, you found that there are 3 unmarked gold bars in the place where you had left your own gold bar. When you measured their mass **using the same scale**, you found that:

gold bar a = 0.991 kg

gold bar b = 1.002 kg

gold bar c = 1.005 kg

Which of these gold bars **could possibly** be your original unmarked gold bar?

- A. None of them
- B. Gold bar b only
- C. Either gold bar a or gold bar b only
- D. Either gold bar b or gold bar c only
- E. Any of them

WORKING

Answer _____

1.2 Motion in a Straight Line

For motion along a line we use **displacement** s , **velocity** $v = \frac{ds}{dt}$ and **acceleration** $a = \frac{dv}{dt}$. Speed is the magnitude of velocity; distance is the total path length.

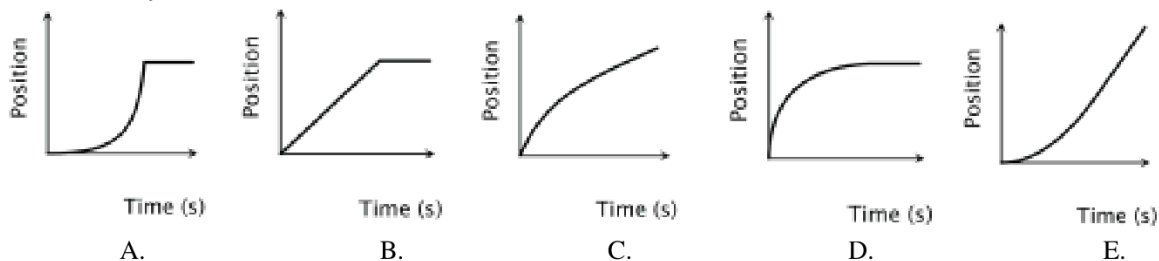
Reading motion graphs

Graph	Gradient gives	Area under graph gives
$s-t$	velocity v	(not used)
$v-t$	acceleration a	displacement s
$a-t$	(rate of change of a)	change in velocity Δv

On an $s-t$ graph, constant positive acceleration from rest shows up as a curve that gets steeper (concave up). Constant velocity is a straight line.

EXAMPLE 1.3 SJPO 2018 Q10

10. The position of an object moving along a linear track is plotted as a function of time. It started from rest and underwent a positive acceleration for some time, followed by a constant velocity. Which of the following graphs correctly shows this situation?



WORKING

Answer _____

Equations of uniformly accelerated motion

If a is **constant**, with initial velocity u and final velocity v after time t :

EQUATIONS OF MOTION (CONSTANT a ONLY)

$$v = u + at \quad s = ut + \frac{1}{2}at^2 \quad s = \frac{1}{2}(u + v)t \quad v^2 = u^2 + 2as$$

Before using them, check your units match. Convert km h^{-1} to m s^{-1} by dividing by 3.6, or keep everything in km and hours.

EXAMPLE 1.4 SJPO 2018 Q13

13. Travelling with an initial speed of 70 km/h, a car accelerates at 6000 km/h^2 along a straight road. How long will it take to reach a speed of 120 km/h?
- A. 30 s
B. 45 s
C. 60 s
D. 70 s
E. 180 s

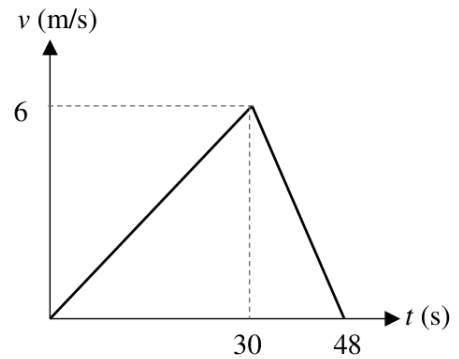
WORKING

Answer _____

EXAMPLE 1.5 SJPO 2018 Q16–17

Q 16 and 17 refer to the following.

A car travels along a straight road with the speed shown by the v-t graph.



16. What is the acceleration of the car from $t=30$ to $t=48$ s?
- A. -54 m/s^2
B. 48 m/s^2
C. -3.0 m/s^2
D. 3.0 m/s^2
E. -0.33 m/s^2
17. What is the total displacement of the car after 48 s?
- A. 36 m
B. 48 m
C. 144 m
D. 180 m
E. 210 m

WORKING

Answer _____

SJPO tip. A "midpoint" question is asking about half the *distance*, not half the *time*. At half the time the speed is the arithmetic mean $\frac{1}{2}(u + v)$. At half the distance, use $v^2 = u^2 + 2as$ twice, which gives the root-mean-square $\sqrt{\frac{1}{2}(u^2 + v^2)}$.

EXAMPLE 1.6 **SJPO 2024 Q3**

3. A particle travels in a straight path with constant acceleration. The particle has velocity u at the start of its path and velocity v at the end of its path. What is the velocity of the particle at the midpoint of its path?

- A. $\frac{1}{2}(v - u)$
- B. $\frac{1}{2}(u + v)$
- C. $\sqrt{uv}$
- D. $\sqrt{u^2 + v^2}$
- E. $\sqrt{\frac{1}{2}(u^2 + v^2)}$

WORKING**Answer** _____**Reaction time and stopping distance**

During the reaction time t_r a driver keeps moving at constant speed. Only then does braking begin:

$$s_{\text{stop}} = u t_r + \frac{u^2}{2|a|}$$

Questions about whether a driver should brake or carry on usually compare s_{stop} with the distance to the line. They also compare the distance covered at constant speed with the time available.

EXAMPLE 1.7 **SJPO 2022 Q1-3**

For questions 1-3.

The road is level, and a driver is driving at the speed limit of 50 km/h (i.e. 14 m/s) when he sees the traffic light ahead turns yellow. He does not want to go over the speed limit and wants to enter the intersection before the light turns red. Assume the magnitude of the car's maximum deceleration rate is 3.0 m/s^2 and the human reaction time is 1.0 s.

For questions 1 and 2: Can he avoid running the red light if the distance to the intersection and the duration of the yellow light are as follows:

1. 50 m and 3 s

- a) Yes, by applying his brake and comes to a stop before the intersection only.
- b) Yes, by continuing to drive at the speed limit only.
- c) Yes, both a) and b) work.
- d) No, both a) and b) do not work, and he will run the red light.

2. 40 m and 2 s

- a) Yes, by continuing to drive at the speed limit only.
- b) Yes, by applying his brake and comes to a stop before the intersection only.
- c) Yes, both a) and b) work.
- d) No, both a) and b) do not work, and he will run the red light.

3. What is the minimum duration for the yellow light so that the law-abiding driver can avoid running the red light, either by continuing to drive at the speed limit or by applying his brake and comes to a stop before the intersection, when he is approaching one?

- a) 6.33s
- b) 2.33s
- c) 4.33s
- d) 3.33s
- e) 5.33s

WORKING

Answer _____

1.3 Relative Motion

Velocity is always measured relative to some observer. If $\vec{v}_{AG}$ is the velocity of A relative to the ground and $\vec{v}_{BG}$ that of B, then the velocity of A **as seen by B** is

RELATIVE VELOCITY

$$\vec{v}_{AB} = \vec{v}_{AG} - \vec{v}_{BG} \quad \text{equivalently} \quad \vec{v}_{AG} = \vec{v}_{AB} + \vec{v}_{BG}$$

Strategy: decide on a ground frame first and work out everyone's ground velocity, then subtract. A stationary object (a tree) seen to move at speed v tells you the observer's own ground speed is v .

EXAMPLE 1.8 SJPO 2024 Q2

2. Chris is seated on a northbound train when he sees a southbound train approaching him with speed u . At the same time, he sees a tree moving away from him at speed v . At what speed does a passenger seated on the southbound train observe this tree to be moving?

- A. $u + v$
- B. $u - v$
- C. $u - 2v$
- D. $2u + v$
- E. $2u - v$

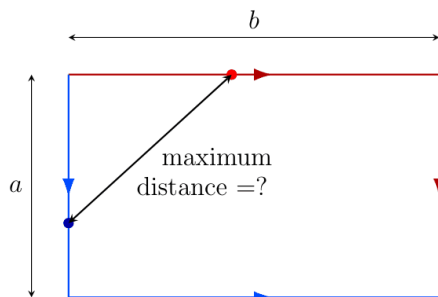
WORKING

Answer _____

For two bodies moving at the *same* speed in the same direction, their separation stays constant. The separation changes only when their velocity components differ. To find a maximum or minimum separation, look for the moment when the relative velocity has no component along the line joining them.

Question 3

Two runners start on the same corner of a rectangular track (e.g. upper-left corner) of sides a and b , with $a < b$. They begin running towards the opposite corner of the track at the same speed through two different paths around the track. What is the maximum distance reached between the two runners?



- (a) a
- (b) b
- (c) $a\sqrt{2}$
- (d) $b\sqrt{2}$
- (e) $\sqrt{a^2 + b^2}$

WORKING

Answer _____

1.4 Projectile Motion

Near Earth's surface, with air resistance neglected, a projectile has constant acceleration $\vec{g}$ downward. Horizontal and vertical motions are **independent**:

PROJECTILE LAUNCHED AT SPEED U , ANGLE θ ABOVE HORIZONTAL

$$x = (u \cos \theta) t \quad y = (u \sin \theta) t - \frac{1}{2}gt^2 \quad v_x = u \cos \theta \text{ (constant)}, \quad v_y = u \sin \theta - gt$$

$$\text{Range on level ground } R = \frac{u^2 \sin 2\theta}{g} \quad \text{Max height } H = \frac{u^2 \sin^2 \theta}{2g}$$

- Speed is *minimum* at the top of the flight, where $v_y = 0$.

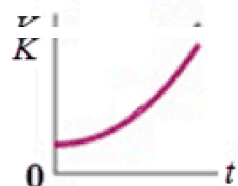
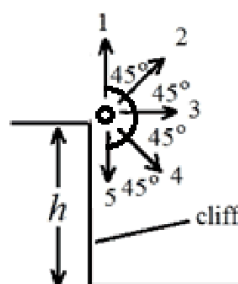
- Because $K = \frac{1}{2}m(v_x^2 + v_y^2)$, $\frac{dK}{dt} = -mg v_y$. So K has zero gradient exactly when the motion is horizontal.

EXAMPLE 1.10 SJPO 2022 Q6–7

For questions 6 and 7.

A ball is launched with a non-zero velocity v_0 in one of the five directions from the edge of a cliff. Direction 1 is vertically up and direction 5 is vertically down, and direction 3 is rightward horizontally while directions 2 (4) are mid-way between 1 and 3 (3 and 5).

The block's kinetic energy K varies with time as shown. Note that the gradient of the K - t graph is **zero** at $t = 0$.



6. Which is the direction of the initial velocity of the ball?

- a) 4
- b) 2
- c) 3
- d) 1
- e) 5

7. Ignoring air resistance, the time taken for the ball to land at the bottom of the cliff is ____.

- a) $\frac{v_0}{g} \left(1 + \sqrt{1 + \frac{2gh}{v_0^2}} \right)$
- b) $\frac{v_0}{\sqrt{2}g} \left(1 + \sqrt{1 + \frac{4gh}{v_0^2}} \right)$
- c) $\sqrt{2gh}$
- d) $\frac{v_0}{\sqrt{2}g} \left(\sqrt{1 + \frac{4gh}{v_0^2}} - 1 \right)$
- e) $\frac{v_0}{g} \left(\sqrt{1 + \frac{2gh}{v_0^2}} - 1 \right)$

WORKING

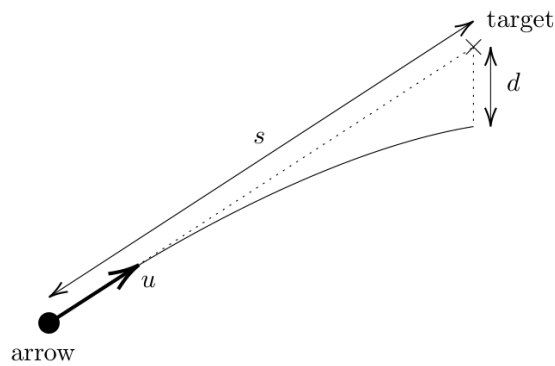
Answer _____

The free-fall frame

In a frame falling freely with acceleration $\vec{g}$, gravity disappears and every projectile moves in a **straight line at constant velocity**. This makes "aiming" problems simple. An arrow aimed straight at a target travels along the line of sight in the falling frame, so in the ground frame it ends up a distance $\frac{1}{2}gt^2$ below the target.

EXAMPLE 1.11 SJPO 2024 Q11

11. Paul fires an arrow at initial speed u towards a target a distance s away from him. He unfortunately makes the mistake of aiming directly at the target, and as a result, his arrow misses the target by a vertical distance d . What is d ?



- A. $\frac{u^2}{2g}$
- B. $\frac{gs^2}{2u^2}$
- C. $\frac{2gs^2}{u^2}$
- D. $\left(\frac{u^2}{2g} - 1\right)s$
- E. $\left(\frac{gs}{2u^2} - 1\right)s$

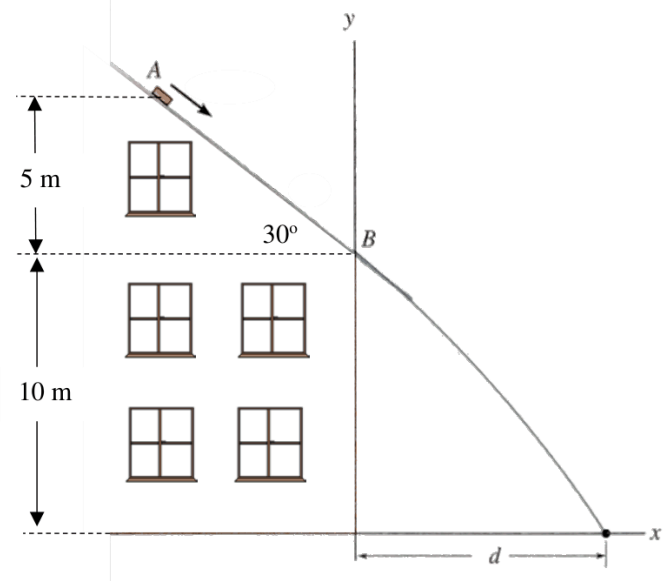
WORKING

Answer _____

EXAMPLE 1.12 SJPO 2018 Q18

18. A 2 kg block slides down a smooth roof which is angled at 30° to the horizontal. When it reaches A, it has a speed of 2.0 m/s. It leaves the surface of the roof at B and strikes the ground at a distance d from the wall of the building. What is the distance d ?

- A. 3.96 m
B. 8.66 m
C. 8.78 m
D. 17.3 m
E. 17.8 m

**WORKING**

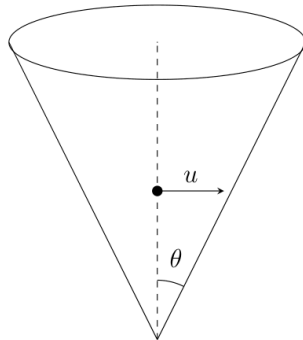
Answer _____

Bouncing projectiles

In an elastic bounce off a smooth surface, the velocity component along the surface is unchanged and the normal component is reversed. If a projectile hits a surface *perpendicularly*, it retraces its path exactly.

Question 13

A particle is held above the tip of an inverted hollow cone of half-angle θ . The particle is projected with a horizontal speed u such that it bounces back and forth indefinitely between the two sides of the cone symmetrically. What is the period of its motion? Neglect air resistance and assume all collisions are elastic.



- (a) $\frac{u}{g} \sin \theta$
- (b) $\frac{2u}{g} \sin \theta$
- (c) $\frac{4u}{g} \sin \theta$
- (d) $\frac{u}{g} \tan \theta$
- (e) $\frac{4u}{g} \tan \theta$

WORKING

Answer _____

2 Newton's Laws and Dynamics

Forces cause accelerations. Free-body diagrams, friction, connected bodies, and the fictitious forces that appear in accelerating frames.

2.1 Newton's Laws and Free-Body Diagrams

2.2 Friction

2.3 Connected Bodies and Pulleys

2.4 Accelerating Frames and Fictitious Forces

LEARNING OUTCOMES

Candidates should be able to:

- state Newton's three laws and identify Newton's third-law pairs correctly
- draw free-body diagrams showing only real forces acting on a chosen body
- apply $F_{\text{net}} = ma$ to single bodies and to systems treated as one object
- use the static and kinetic friction models $f_s \leq \mu_s N$ and $f_k = \mu_k N$
- solve connected-particle problems involving strings, pulleys and stacked blocks
- analyse motion in an accelerating frame using a fictitious force $-m\vec{a}_{\text{frame}}$

2.1 Newton's Laws and Free-Body Diagrams

Newton's first law. A body stays at rest or moves with constant velocity unless a resultant force acts on it.

Newton's second law. The resultant force equals the rate of change of momentum: $\vec{F}_{\text{net}} = \frac{d\vec{p}}{dt}$, which becomes $\vec{F}_{\text{net}} = m\vec{a}$ for constant mass.

Newton's third law. If A exerts a force on B, then B exerts a force on A that is equal in magnitude, opposite in direction and *of the same type*.

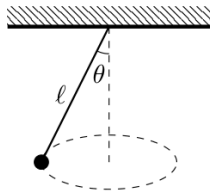
A third-law pair always acts on **two different bodies**, so the two forces never appear on the same free-body diagram. The weight of a book and the normal force on it are *not* a third-law pair. The partner of the book's weight is the book's gravitational pull on the Earth.

Free-body diagrams (FBDs)

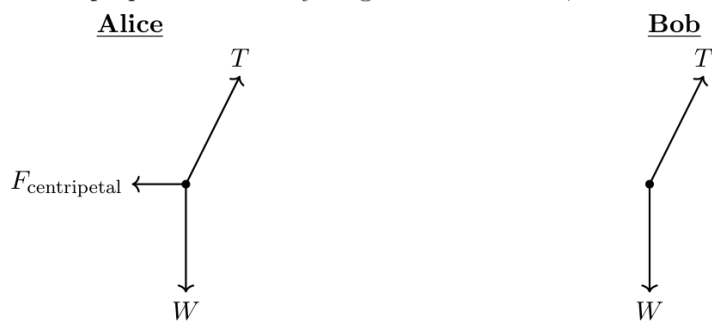
- Isolate one body and draw only the **real** forces acting on it: weight, normal contact, tension, friction, drag, spring force and so on.
- "Centripetal force" is not a new kind of force. It is the *name* for the resultant force that points towards the centre. Never add it as an extra arrow.
- Choose axes along the acceleration if you know its direction.

EXAMPLE 2.1 **SJPO 2024 Q7**

7. In a “circular pendulum”, a mass m is suspended by a string of length ℓ and travels in a circular path at constant speed.



Alice and Bob each propose a free-body diagram for the mass, as shown below.



Whose free-body diagram is correct, and why?

- A. Alice, as the mass must be in equilibrium.
- B. Alice, as the centripetal force is a fictitious force that must be included.
- C. Bob, as tension and weight are the only forces on the mass.
- D. Bob, as tension and weight form an action-reaction pair.
- E. Both are valid diagrams depending on the frame of reference chosen.

WORKING

Answer _____

The system method

Bodies that move together with the same acceleration can be treated as one object of mass M_{total} . Internal forces cancel. Find a from the external forces, then isolate one body to find an internal force.

SYSTEM METHOD

$$a = \frac{\sum F_{\text{external}}}{\sum m} \quad \text{then for one part: } F_{\text{on part}} = m_{\text{part}} a$$

EXAMPLE 2.2 **SJPO 2024 Q4**

4. n identical blocks (where $n > 2$) are placed side-by-side in a horizontal row on frictionless ground. The combined row of blocks is pushed from the left with a force F . What is the force between the two rightmost blocks?

- A. F
- B. nF
- C. $\frac{F}{n}$
- D. $\frac{F}{n-1}$
- E. $\frac{(n-1)F}{n}$

WORKING**Answer** _____**Terminal velocity**

A body falling through a fluid with drag $F_{\text{drag}}(v)$ speeds up until the drag balances its weight (minus buoyancy). At terminal velocity $a = 0$. For linear drag $F = cv$, this gives $v_T = mg/c$.

EXAMPLE 2.3 **SJPO 2018 Q2**

2. A tiny spherical raindrop of diameter $D = 0.1$ mm experiences a linear drag force while falling at speed v , given by $F_{\text{drag}} = cv$, where $c = 1.55 \times 10^{-6}$ N s m⁻¹. What is its terminal speed?

- A. 1.7 mm / s
- B. 3.3 mm / s
- C. 8.5 mm / s
- D. 26 mm / s
- E. 33 mm / s

WORKING**Answer** _____**2.2 Friction**

Friction acts along the contact surface and opposes **relative motion**, or the tendency towards it, between the two surfaces.

FRICTION MODEL

static: $f_s \leq \mu_s N$ (adjusts to whatever is needed) kinetic: $f_k = \mu_k N$ (constant), $\mu_k < \mu_s$

- Static friction is only as large as it needs to be. Do not write $f = \mu_s N$ unless the surfaces are about to slip.
- On an incline of angle θ , a block is on the point of slipping when $\tan \theta = \mu_s$. Once moving, $a = g(\sin \theta - \mu_k \cos \theta)$.
- A useful identity: if $\tan \theta = \mu$, then $\sin \theta = \frac{\mu}{\sqrt{1 + \mu^2}}$ and $\cos \theta = \frac{1}{\sqrt{1 + \mu^2}}$.

EXAMPLE 2.4 SJPO 2026 Q6

Question 6

A block is released from rest on an inclined ramp, and slides down the ramp with acceleration $a > 0$. The ramp has coefficients of static and kinetic friction μ_s and μ_k respectively. What is the smallest possible value of a ?

- (a) $(1 - \mu_k)g$
- (b) $(\mu_s - \mu_k)g$
- (c) $\frac{\mu_s}{\sqrt{1 + \mu_s^2}}g$
- (d) $\frac{\mu_k}{\sqrt{1 + \mu_s^2}}g$
- (e) $\frac{\mu_s - \mu_k}{\sqrt{1 + \mu_s^2}}g$

WORKING

Answer _____

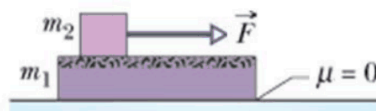
Stacked blocks: do they move together?

1. **Assume** they move together and find the common acceleration from the system.
2. Isolate the block that is *not* directly pushed or pulled. Find the friction it needs.
3. Check: is the friction needed $\leq \mu_s N$? If yes, the assumption holds. If not, they slip, so use $\mu_k N$ on each block separately.

EXAMPLE 2.5 SJPO 2022 Q12–13

For questions 12 and 13

12. A slab of mass m_1 rests on a frictionless floor, and a block of mass m_2 rests on top of the slab. Between the block and slab, the coefficient of static friction is μ_s and the coefficient of kinetic friction is μ_k . A horizontal force of magnitude F begins to pull directly on the block, as shown. The condition for the two masses to move as one piece is, _____, where $M = m_1 + m_2$.



a) $\mu_s \leq \frac{m_1}{m_2} \frac{F}{Mg}$

b) $\mu_s \leq \frac{m_2}{m_1} \frac{F}{Mg}$

c) $\mu_s \geq \frac{m_1}{m_2} \frac{F}{Mg}$

d) $\mu_s \geq \frac{m_2}{m_1} \frac{F}{Mg}$

e) The two masses will always move as one piece since there is friction between them and no friction between the slab and the floor.

Hint: The kinetic friction on an object is given by $\mu_k N$ where N is the normal force on the object. The maximum static friction on an object is given by $\mu_s N$.

13. Suppose $m_1 = 50$ kg and $m_2 = 20$ kg, $\mu_s = 0.55$, $\mu_k = 0.40$ and $F = 140$ N, what is the resulting acceleration (in $\text{m} \cdot \text{s}^{-2}$) of the block? Take $g = 10$ N/kg.

a) 7.0

b) 3.0

c) 3.3

d) 1.6

e) 2.0

WORKING

2.3 Connected Bodies and Pulleys

For an **inextensible, light string** over a **light, frictionless pulley**:

- the tension is the same all along the string;
- connected bodies have accelerations of equal magnitude (along the string).

Treat the string-connected bodies as one system moving "along the string". The driving force is whatever gravity component acts along the string direction, minus any resistances along it:

BLOCK ON SMOOTH TABLE, HANGING BLOCK M_2

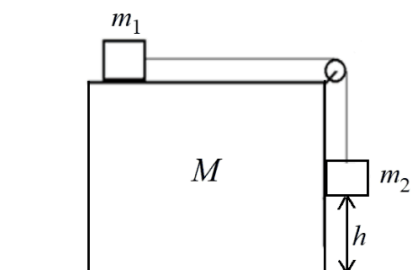
$$a = \frac{m_2 g}{m_1 + m_2} \quad T = m_1 a = \frac{m_1 m_2}{m_1 + m_2} g$$

After finding a , use the equations of motion. For example, a block falling a height h from rest takes $t = \sqrt{2h/a}$.

EXAMPLE 2.6 SJPO 2022 Q17–18

For questions 17– 21

A block of mass m_1 is placed on top of a block of mass M . Holding block m_1 stationary, it is joined by a string which passes over the pulley to a block of mass m_2 hanging by the side of (and in contact with) block M , and is at height h above the floor. We will assume that the string is inextensible and massless, and all the contacts between objects in relative motion are frictionless. Furthermore, we will be concerned with the part of the motion before m_1 reaches the pulley.



For questions 17 and 18, assume that M is fixed to the floor, and the pulley is massless. After m_1 is released from rest, both m_1 and m_2 move smoothly with a constant acceleration and the string remains taut.

17. What is the acceleration of blocks m_1 and m_2 ?

- a) $\frac{m_1}{m_1+m_2}g$
- b) $\frac{m_2}{m_1+m_2}g$
- c) g
- d) $\frac{m_1+m_2}{m_1}g$
- e) $\frac{m_1+m_2}{m_2}g$

18. What is the time taken for m_2 to reach the floor after m_1 is released from rest?

- a) $\sqrt{\frac{2h(m_1+m_2)}{m_1g}}$
- b) $\sqrt{\frac{2h(m_1+m_2)}{m_2g}}$
- c) $\sqrt{\frac{2h}{g}}$
- d) $\sqrt{\frac{2h m_1}{(m_1+m_2)g}}$
- e) $\sqrt{\frac{2h m_2}{(m_1+m_2)g}}$

WORKING

Answer _____

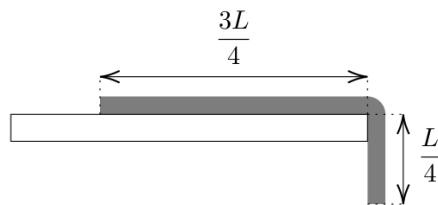
Chains and ropes with mass

A uniform chain partly hanging over an edge is a "connected system" too. Its total mass m accelerates, while only the hanging part's weight drives it. If a fraction f hangs over a smooth edge, the initial acceleration is $a = fg$. With friction on the part lying on the table, the chain can only start moving if the hanging weight exceeds $\mu_s \times$ (normal force on the part on the table).

EXAMPLE 2.7 SJPO 2024 Q5–6

The following information applies to both questions 5 and 6.

A uniform chain of length L is placed on a table, with a length $\frac{L}{4}$ hanging over the edge of the table. When the chain is released from rest, it begins sliding off the table.



5. Suppose that the tabletop is frictionless. What is the initial acceleration of the chain?
- A. $\frac{g}{4}$
 - B. $\frac{g}{3}$
 - C. $\frac{2g}{3}$
 - D. $\frac{3g}{4}$
 - E. g
6. Suppose now that the tabletop has friction. Given that the chain still slides off the table upon release, what is the maximum possible coefficient of static friction of the tabletop?
- A. $\frac{1}{4}$
 - B. $\frac{1}{3}$
 - C. $\frac{2}{3}$
 - D. $\frac{3}{4}$
 - E. 1

WORKING

Answer _____

2.4 Accelerating Frames and Fictitious Forces

Newton's laws hold in **inertial** frames. In a frame accelerating at $\vec{a}_0$, such as a train, bus or lift, we can still use $\vec{F} = m\vec{a}_{\text{rel}}$ if we add a **fictitious force** on every body:

FICTITIOUS (PSEUDO) FORCE

$$\vec{F}_{\text{fict}} = -m\vec{a}_0 \quad \implies \quad \vec{g}_{\text{eff}} = \vec{g} - \vec{a}_0$$

In the accelerating frame everything behaves as if gravity were $\vec{g}_{\text{eff}}$, of magnitude $\sqrt{g^2 + a_0^2}$ for horizontal $\vec{a}_0$, tilted **backwards**, away from the acceleration.

- A plumb line hangs along $\vec{g}_{\text{eff}}$, at angle $\tan^{-1}(a_0/g)$ to the vertical.
- A liquid surface settles **perpendicular** to $\vec{g}_{\text{eff}}$, so it is tilted at $\tan \theta = a_0/g$, with the liquid piled up at the back.

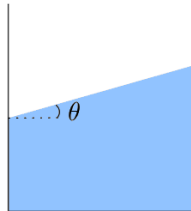
EXAMPLE 2.8 SJPO 2018 Q3

3. A plumb line is held steady while being carried in a moving train. The mass of the plumb bob is m , and the train is accelerating in the forward direction at $0.5g$. What is the tension in the string?
- A. $0.5\,mg$
B. $1.12\,mg$
C. $1.22\,mg$
D. $2\,mg$
E. Cannot be determined from the given information.

WORKING**Answer** _____

EXAMPLE 2.9 SJPO 2024 Q8

8. On board a shaky bus, Galen observes a glass of water and notices its surface to be tilted towards the right by an angle $\theta = 35^\circ$. What is the acceleration of the bus, and in which direction?



- A. 5.6 m s^{-2} towards the left
- B. 5.6 m s^{-2} towards the right
- C. 6.9 m s^{-2} towards the left
- D. 6.9 m s^{-2} towards the right
- E. 9.8 m s^{-2} downwards

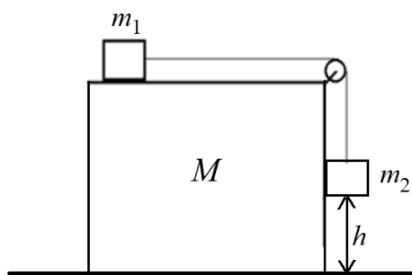
WORKING**Answer** _____

Which frame? Ground frame: draw only real forces, and the net force equals $m\vec{a}$. Accelerating frame: real forces plus $-m\vec{a}_0$, and the body may be at rest. Either works, but never mix them. The 2024 Q7 trap (Section 2.1) is exactly this: an outward centrifugal force exists only in the rotating frame.

EXAMPLE 2.10 SJPO 2022 Q20

For questions 17- 21

A block of mass m_1 is placed on top of a block of mass M . Holding block m_1 stationary, it is joined by a string which passes over the pulley to a block of mass m_2 hanging by the side of (and in contact with) block M , and is at height h above the floor. We will assume that the string is inextensible and massless, and all the contacts between objects in relative motion are frictionless. Furthermore, we will be concerned with the part of the motion before m_1 reaches the pulley.



For questions 19 and 20, all assumptions remain except that M is no longer fixed to the floor but can move without friction on the floor.

20. By applying a constant rightward force on M , both m_1 and m_2 can remain at rest relative to M without having to hold them. This force is ____.

- a) $\frac{m_1}{(M+m_1+m_2)m_2}g$
- b) $\frac{m_2}{(M+m_1+m_2)m_1}g$
- c) $\frac{(M+m_1+m_2)m_1}{m_2}g$
- d) $\frac{(M+m_1+m_2)m_2}{m_1}g$
- e) none of the above

WORKING

Answer _____

3 Forces in Equilibrium

Statics: when nothing accelerates and nothing turns. Resolving forces, moments, centre of mass, and the moment a support goes slack or a body tips.

3.1 Equilibrium of Concurrent Forces

3.2 Moments and Centre of Mass

LEARNING OUTCOMES

Candidates should be able to:

- state the two conditions for equilibrium of a rigid body
- resolve forces and solve for unknowns using components or a closed vector triangle
- calculate moments and apply the principle of moments about a well-chosen pivot
- locate the centre of mass of composite bodies
- find the condition for a support to go slack or a body to tip about an edge

3.1 Equilibrium of Concurrent Forces

A body is in **translational equilibrium** when the resultant force on it is zero:

TRANSLATIONAL EQUILIBRIUM

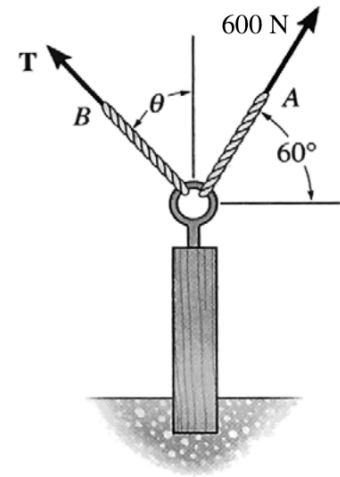
$$\sum F_x = 0, \quad \sum F_y = 0 \quad (\text{three coplanar forces: they form a closed triangle})$$

- Resolve every force along two perpendicular axes. Choose axes that line up with as many forces as possible.
- A **normal force** is perpendicular to the contact surface. On a curved surface $y = f(x)$, the surface makes angle ϕ with the horizontal where $\tan \phi = \frac{dy}{dx}$. The normal force makes the same angle ϕ with the vertical.
- Smooth spheres in contact push on each other along the **line joining their centres**.

EXAMPLE 3.1 SJPO 2018 Q12

12. A wooden peg is to be pulled out of the ground using two ropes A and B. Rope A is subject to a force of 600 N at 60° to the horizontal. Rope B is pulled at a fixed angle θ to the vertical. If the resultant force acting on the post is to be 1600 N vertically, what should be the force T on rope B?

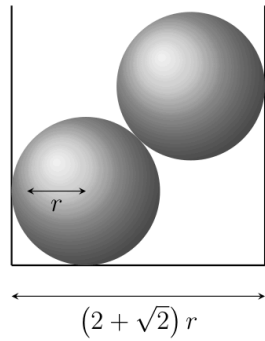
- A. 1121 N
- B. 1334 N
- C. 1400 N
- D. 1924 N
- E. 2040 N

**WORKING**

Answer _____

Question 11

A box of base width $(2 + \sqrt{2})r$ is placed on a horizontal ground. Two identical spheres of mass m and radius r are packed inside it. What is the force exerted between the two spheres? Assume all surfaces are smooth.

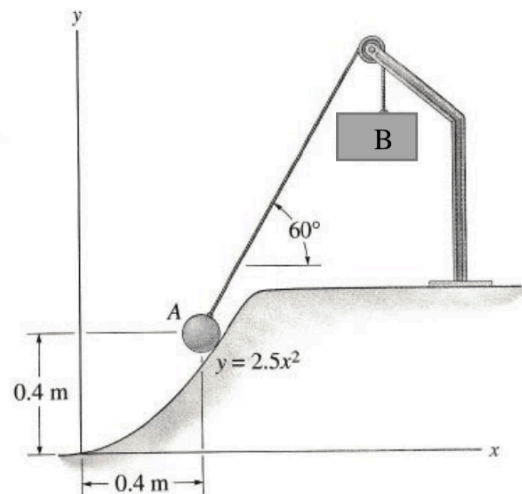


- (a) mg
- (b) $2mg$
- (c) $\frac{mg}{2}$
- (d) $mg\sqrt{2}$
- (e) $\frac{\sqrt{2}}{2}mg$

WORKING**Answer** _____

EXAMPLE 3.3 SJPO 2018 Q6

6. A 4 kg sphere rests on the smooth parabolic surface which is given by the equation $y = 2.5x^2$. The sphere just touches the surface at $(x, y) = (0.4 \text{ m}, 0.4 \text{ m})$. Determine the mass m_B of block B needed to hold the sphere in equilibrium.
- A. 4.14 kg
B. 3.75 kg
C. 3.58 kg
D. 2.93 kg
E. 2.14 kg



WORKING

Answer _____

3.2 Moments and Centre of Mass

Moment (torque) of a force about a point = force $\times$ perpendicular distance from the point to the line of action of the force: $\tau = Fd_{\perp} = Fr \sin \theta$.

CONDITIONS FOR EQUILIBRIUM OF A RIGID BODY

$$\sum \vec{F} = 0 \quad \text{and} \quad \sum \tau = 0 \text{ about any point}$$

Choose the pivot wisely. Take moments about the point where the unknown force you *don't* care about acts, because that force then has no moment.

Centre of mass

The weight of a body acts at its centre of mass (CM). For a collection of parts:

$$x_{\text{cm}} = \frac{\sum m_i x_i}{\sum m_i}$$

For a uniform rod, the CM is at its midpoint. Use symmetry wherever you can.

EXAMPLE 3.4 SJPO 2018 Q5

5. A thin wire is bent into the form of a three-sided shape as shown below. Each segment has equal length l . The height of the centre of mass from the bottom of the shape is _____.

- A. $l/2$
- B. $2l/3$
- C. $l/3$
- D. $l/4$
- E. $2l/5$



WORKING

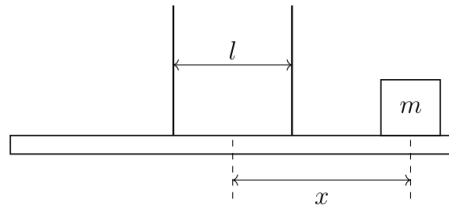
Answer _____

Going slack and tipping

- A support (string, prop or edge) **goes slack** or loses contact when the force it must exert drops to zero. At that instant, the CM of the whole system lies directly above or below the remaining support.
- A body on the point of **tipping** about an edge carries all its contact force at that edge. Take moments about the edge.
- To find the minimum force needed to topple something, apply it so that it has the **largest possible moment arm**.

Question 9

A long uniform rod of mass M is held horizontal by two strings, attached symmetrically to the rod at distance l apart. A small block of mass m initially rests on the rod's centre. It is then slowly moved towards the right. At what distance x between the block and the rod's centre does one of the strings start to become slack?



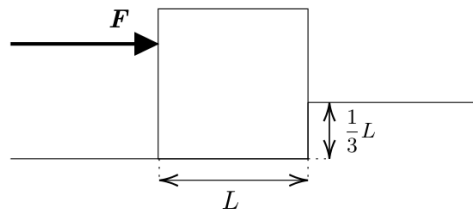
- (a) $\frac{l}{2} \frac{M+m}{m}$
- (b) $\frac{l}{2} \frac{M-m}{m}$
- (c) $\frac{l}{2} \frac{M+m}{M}$
- (d) $\frac{l}{2} \frac{m}{M+m}$
- (e) $\frac{l}{2} \frac{M}{M+m}$

WORKING

Answer _____

EXAMPLE 3.6 **SJPO 2024 Q9**

9. A tired delivery worker has a cube-shaped box of mass M and side length L on the ground, beside a step of height $\frac{1}{3}L$. In an attempt to bring the box up the step, he applies a horizontal force F on the box. What is the smallest value of F required for him to succeed?



- A. Mg
- B. $\frac{1}{2}Mg$
- C. $\frac{2}{3}Mg$
- D. $\frac{3}{4}Mg$
- E. $\frac{3}{2}Mg$

WORKING**Answer** _____

4 Work, Energy and Power

Energy methods skip the details of the motion: track where the energy goes, and the speed, height or compression falls out.

4.1 Work and the Work–Energy Theorem

4.2 Conservation of Energy and Dissipation

LEARNING OUTCOMES

Candidates should be able to:

- calculate work done by constant forces, and by a spring as the area under a force–extension graph
- apply the work–energy theorem $W_{\text{net}} = \Delta K$
- use gravitational and elastic potential energy, and conservation of mechanical energy
- account for work done by friction and other resistive forces as energy dissipated
- relate power to force and velocity, $P = Fv$

4.1 Work and the Work–Energy Theorem

Work done by a constant force $\vec{F}$ over a displacement $\vec{s}$: $W = \vec{F} \cdot \vec{s} = Fs \cos \theta$. For a variable force, $W = \int F dx$, which is the area under the F - x graph.

WORK-ENERGY THEOREM

$$W_{\text{net}} = \Delta K = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

Springs

A Hookean spring exerts $F = -kx$, where x is the extension from natural length. Its elastic potential energy is $U = \frac{1}{2}kx^2$. The work done **by the spring** as the extension changes from x_1 to x_2 is

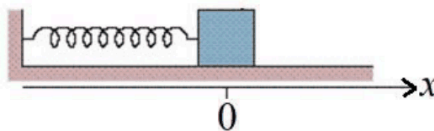
$$W_{\text{spring}} = -\Delta U = \frac{1}{2}k(x_1^2 - x_2^2)$$

Positive work by the spring means its stored energy decreased. Only x^2 matters, so a compression of 3 cm stores the same energy as an extension of 3 cm.

EXAMPLE 4.1 SJPO 2022 Q8

For questions 8 and 9.

A spring that obeys Hooke's law and a block of mass 2.0 kg are set up as shown. When the block is at $x = 0$, the spring is at its natural length. The block is pulled out to $x = 5.0$ cm, and a force of magnitude 10 N is required to hold it there. Then the block is released. Ignore friction.



8. How much is the work done by the spring on the block to move it from $x = 5.0$ cm to $x = -3.0$ cm?

- a) 0.16J
- b) -0.16J
- c) 0.08J
- d) -0.08J
- e) cannot be determined as the spring constant is not given

WORKING

Answer _____

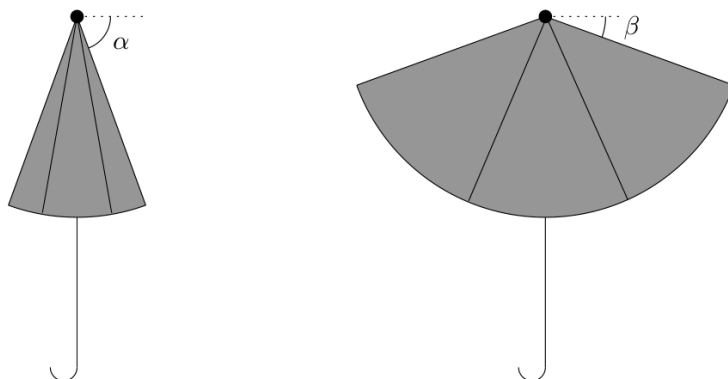
Gravitational potential energy

Near the surface, $\Delta U = mg \Delta h$. For an extended body, h is the height of its **centre of mass**, so lifting a uniform rod hinged at one end raises its CM by $\frac{1}{2}l \Delta(\sin \theta)$.

The work you must do to change a configuration *slowly*, without giving it kinetic energy, equals the increase in potential energy.

EXAMPLE 4.2 SJPO 2024 Q13

13. The frame of an umbrella is constructed by attaching 6 uniform rods, each of mass $m = 0.05$ kg and length $l = 1.0$ m, to a common centre. The frame makes an angle $\alpha = 80^\circ$ to the horizontal when the umbrella is closed, and an angle $\beta = 20^\circ$ when the umbrella is open, as drawn below. (Note: Not all 6 rods are drawn due to perspective.)



How much work needs to be done to open the umbrella? Assume that the umbrella's central axis is kept vertical and stationary, and that the mass of its fabric is negligible compared to the mass of its frame.

- A. 0.95 J
- B. 1.1 J
- C. 1.9 J
- D. 2.3 J
- E. 2.9 J

WORKING**Answer** _____**Power**

$$P = \frac{dW}{dt} = Fv \quad \text{efficiency} = \frac{\text{useful power out}}{\text{power in}}$$

4.2 Conservation of Energy and Dissipation

If only **conservative** forces (gravity, springs) do work, mechanical energy is conserved:

ENERGY CONSERVATION

$$K_i + U_i = K_f + U_f \quad \text{in general:} \quad K_i + U_i + W_{\text{non-cons}} = K_f + U_f$$

Work by friction and drag ($W_{\text{non-cons}} < 0$) is dissipated as thermal energy. For kinetic friction on a level surface, energy dissipated $= \mu_k mg d$.

Turning points

At the lowest point of a bungee jump, or the maximum compression of a spring, the body is momentarily at rest, so $K = 0$ there. Measure gravitational PE from that lowest point to make the algebra short.

EXAMPLE 4.3 SJPO 2018 Q19

19. **Two** adventurous Physics students, each weighing 60 kg, jump off a 43 m high bridge on a Bungee cord. The length of the cord is such that the students together will *just* touch the water and rebound. While Bungee cords become softer (more elastic) with increasing extension, for our calculations we can approximate the cord as having a constant stiffness of $k = 330 \text{ N/m}$, and we can ignore the height of the students and where the cord is tied on their body. What is the unstretched length of the cord?
- A. 10.0 m
 - B. 17.5 m
 - C. 25.5 m
 - D. 32.5 m
 - E. 40.0 m

WORKING

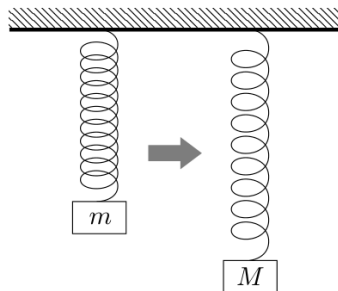
Answer _____

Where did the energy go?

When a mass is added gently to a spring and released, it oscillates and eventually settles, usually lower. The resistive forces dissipate exactly the difference between the initial and final **total potential energy** (gravitational plus elastic), because the body starts and ends at rest.

EXAMPLE 4.4 SJPO 2024 Q15

15. A mass m is suspended at rest by an ideal spring of force constant k . At this position, an extra mass is gently added so that the total mass is now M , before it is released from rest. The spring-mass system subsequently oscillates, but eventually settles to a rest due to damping by resistive forces. How much work is done in total by these resistive forces?



- A. $-\frac{M^2 g^2}{2k}$
B. $-\frac{M(M-m)g^2}{2k}$
C. $-\frac{(M^2 - m^2)g^2}{2k}$
D. $-\frac{(M^2 + m^2)g^2}{2k}$
E. $-\frac{(M-m)^2 g^2}{2k}$

WORKING**Answer** _____

5 Momentum and Collisions

When forces are internal or brief, momentum is the quantity to track: impulse, collisions, restitution and the motion of the centre of mass.

5.1 Impulse and Momentum

5.3 Centre of Mass Motion

5.2 Conservation of Momentum and Collisions

LEARNING OUTCOMES

Candidates should be able to:

- relate impulse to the area under a force–time graph and to the change in momentum
- apply conservation of linear momentum to collisions, explosions and throwing problems
- use the coefficient of restitution, and distinguish elastic from inelastic collisions
- combine momentum and energy conservation, including the common-velocity condition for maximum compression
- use the fact that the centre of mass of an isolated system moves at constant velocity

5.1 Impulse and Momentum

Linear momentum $\vec{p} = m\vec{v}$. **Impulse** of a force $\vec{J} = \int \vec{F} dt$, which is the area under the F - t graph.

IMPULSE–MOMENTUM THEOREM

$$\vec{J} = \int \vec{F} dt = \Delta\vec{p} \quad (\text{constant force: } J = F \Delta t)$$

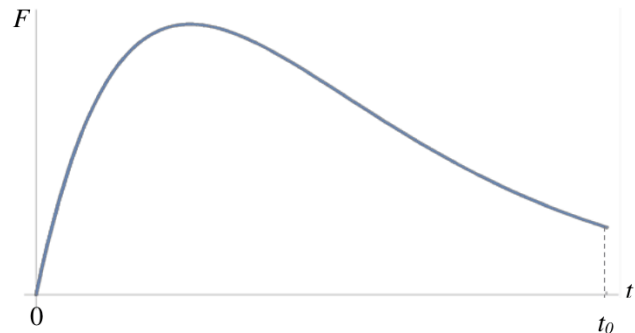
For a non-uniform force given as a formula, integrate it. The integral hint SJPO provides is usually there to be used after a substitution such as $x = t/T$.

EXAMPLE 5.1 **SJPO 2018 Q15**

15. The twitch in an arm muscle develops a force F which can be measured as a function of time t , given by $F = F_0 \left(\frac{t}{T} \right) e^{-t/T}$, where F_0 and T are constants. The function is plotted in the figure. If the ratio $t_0/T = 1$, what is the impulse developed between time $t = 0$ and time $t = t_0$? (Hint: Make use of this integral.

$$\int_0^a x e^{-x} dx = 1 - (1+a)e^{-a} .)$$

- A. $F_0(T(1/e) + t_0(1/e))$
B. $F_0(T(1/e) - t_0(1/e))$
C. $F_0(T(1 - 1/e) + t_0(1/e))$
D. $F_0(T(1 - 1/e) - t_0(1/e))$
E. Indeterminate

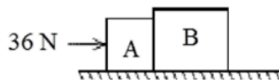
**WORKING****Answer** _____

When a force pushes a group of bodies together, the impulse goes into the momentum of the *whole group*. Each body then gets a share in proportion to its mass, since they share a common velocity.

EXAMPLE 5.2 SJPO 2022 Q4–5

For questions 4 and 5.

Two blocks, A and B, of masses $m_A = 4.0 \text{ kg}$ and $m_B = 20.0 \text{ kg}$ are in contact and are moving with velocity 1.0 ms^{-1} on a horizontal floor. The 36-N constant force is then applied to A for 5 seconds as shown. Ignore friction.



4. After 5 seconds, the momentum (in $\text{kg} \cdot \text{m} \cdot \text{s}^{-1}$) of block B is ____.

- a) 170
- b) 204
- c) 120
- d) 150
- e) 240

5. After 5 seconds, the net work done (in J) on the block A is ____.

- a) 712.5
- b) 142.5
- c) 663
- d) 112.5
- e) 855

WORKING

Answer _____

5.2 Conservation of Momentum and Collisions

CONSERVATION OF LINEAR MOMENTUM

$$\text{If } \sum \vec{F}_{\text{ext}} = 0 : \quad \sum m_i \vec{u}_i = \sum m_i \vec{v}_i$$

During a brief collision, external forces such as gravity and friction are usually negligible compared with the huge internal forces. So momentum is conserved *across* the collision, even if not before or after.

Coefficient of restitution

$$e = \frac{\text{speed of separation}}{\text{speed of approach}} = \frac{v_2 - v_1}{u_1 - u_2}$$

Type	e	Kinetic energy
Perfectly elastic	1	conserved
Inelastic	$0 < e < 1$	some lost
Perfectly inelastic (stick together)	0	maximum loss consistent with momentum

EXAMPLE 5.3 SJPO 2018 Q14

14. Two blocks move towards each other on a smooth table with the velocities as shown. Block A has mass 5 kg and moves at 2 m/s to the right. Block B has mass 2 kg and moves at 5 m/s to the left. The coefficient of restitution e is the ratio of the final relative speed of separation to the initial relative speed of approach of two colliding objects. Take rightwards as positive. If $e = 0.5$, what is the velocity of block B after impact?

- A. - 2.5 m/s
- B. - 1.0 m/s
- C. 0 m/s
- D. 1.0 m/s
- E. 2.5 m/s



WORKING

Answer _____

EXAMPLE 5.4 SJPO 2024 Q12

12. Bobo and Nana are initially at rest on icy frictionless ground. Bobo has mass M_1 and Nana has mass M_2 . Bobo grabs a snowball of mass m and throws it at Nana, who catches it. Bobo and Nana end up moving apart at speeds v_1 and v_2 respectively. What is the ratio $\frac{v_2}{v_1}$?

- A. $\frac{M_1}{M_2}$
B. $\frac{M_2}{M_1}$
C. $\frac{M_1}{M_2 + m}$
D. $\frac{M_2}{M_1 + m}$
E. $\frac{M_1 + m}{M_2 + m}$

WORKING

Answer _____

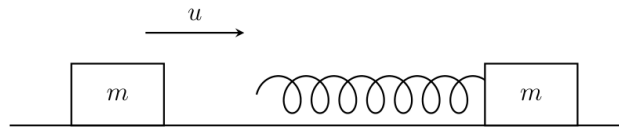
Maximum compression: the common-velocity condition

When two bodies interact through a spring, the spring is most compressed when the bodies have **zero relative velocity**, so they move with a common velocity $v = \frac{\sum m_i u_i}{\sum m_i}$. Then energy conservation gives the compression: $\frac{1}{2} k x_{\max}^2 = K_{\text{initial}} - K_{\text{common}}$.

SJPO tip. For an elastic collision between *equal* masses with one initially at rest, the two final velocities are **perpendicular** (unless the collision is head-on). This follows from $\vec{u} = \vec{v}_1 + \vec{v}_2$ and $u^2 = v_1^2 + v_2^2$, which together give $\vec{v}_1 \cdot \vec{v}_2 = 0$.

EXAMPLE 5.5 SJPO 2026 Q7**Question 7**

A block of mass m is launched with speed u towards an ideal spring of force constant k , which is attached to another block of equal mass m . Both blocks are free to slide on a frictionless horizontal surface. What is the maximum compression of the spring?



- (a) $\frac{u}{2} \sqrt{\frac{m}{k}}$
- (b) $\frac{u\sqrt{2}}{2} \sqrt{\frac{m}{k}}$
- (c) $\frac{u\sqrt{3}}{2} \sqrt{\frac{m}{k}}$
- (d) $\frac{u\sqrt{6}}{2} \sqrt{\frac{m}{k}}$
- (e) $u \sqrt{\frac{m}{k}}$

WORKING

Answer _____

5.3 Centre of Mass Motion

The centre of mass of a system moves as if all the mass were concentrated there and all the external forces acted there:

CENTRE OF MASS

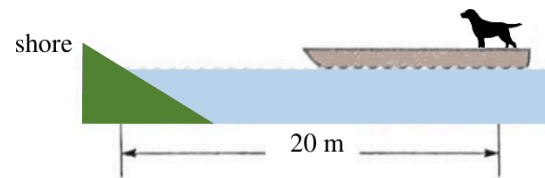
$$M\vec{a}_{\text{cm}} = \sum \vec{F}_{\text{ext}} \quad \Rightarrow \quad \text{if } \sum \vec{F}_{\text{ext}} = 0 \text{ and the system starts at rest, the CM stays put: } \sum m_i \Delta x_i = 0$$

This is the fastest way to handle "person walks on a boat" problems. Write every displacement **relative to the ground**. Then relate them through the relative displacement given in the question.

EXAMPLE 5.6 **SJPO 2018 Q20**

20. A dog weighing 10 kg is standing on a flatboat so that he is 20 m from the shore. The boat weighs 40 kg and has uniform mass distribution. The dog walks 8.0 m on the boat towards the shore and stops. For this calculation, one can assume no friction or drag between the boat and the water. How far is the dog from the shore now?

- A. 12.0 m
B. 13.6 m
C. 14.0 m
D. 16.8 m
E. 29.8 m



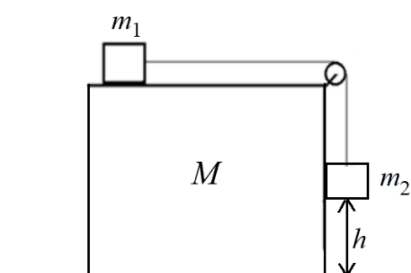
WORKING

Answer _____

EXAMPLE 5.7 SJPO 2022 Q19

For questions 17- 21

A block of mass m_1 is placed on top of a block of mass M . Holding block m_1 stationary, it is joined by a string which passes over the pulley to a block of mass m_2 hanging by the side of (and in contact with) block M , and is at height h above the floor. We will assume that the string is inextensible and massless, and all the contacts between objects in relative motion are frictionless. Furthermore, we will be concerned with the part of the motion before m_1 reaches the pulley.



For questions 19 and 20, all assumptions remain except that M is no longer fixed to the floor but can move without friction on the floor.

19. By considering the motion (or non-motion) of the center of mass and/or otherwise, when m_2 reaches the floor, what is the distance moved by M ? State in which direction if it has moved.

- a) 0
- b) $\frac{m_1 h}{M+m_1+m_2}$, leftward
- c) $\frac{m_1 h}{M+m_1+m_2}$, rightward
- d) $\frac{m_1 h}{M+m_2}$, leftward
- e) $\frac{m_1 h}{M+m_2}$, rightward

WORKING

Answer _____

6 Circular Motion

Moving in a circle means accelerating towards the centre. Find which real forces supply that acceleration, whether the circle is flat, banked or vertical.

6.1 Uniform Circular Motion in a Horizontal Plane

6.2 Vertical and Non-Uniform Circular Motion

LEARNING OUTCOMES

Candidates should be able to:

- use angular displacement, angular velocity and the relation $v = r\omega$
- state that uniform circular motion requires a centripetal acceleration $v^2/r = r\omega^2$ supplied by real forces
- analyse horizontal circles including turntables, banked surfaces and cones
- combine energy conservation with the radial equation for vertical circles
- include tangential acceleration in non-uniform circular motion

6.1 Uniform Circular Motion in a Horizontal Plane

For motion on a circle of radius r : angular velocity $\omega = \frac{d\theta}{dt} = \frac{2\pi}{T}$ and speed $v = r\omega$.

CENTRIPETAL ACCELERATION

$$a_c = \frac{v^2}{r} = r\omega^2 \quad \text{towards the centre} \quad F_{\text{net, radial}} = \frac{mv^2}{r}$$

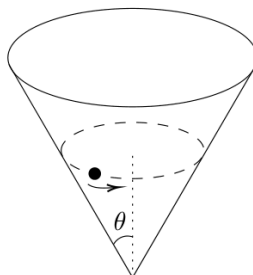
- Identify which real forces (friction, tension, normal force, gravity) have a component towards the centre. That component **is** the centripetal force.
- Turntables / flat bends:** static friction supplies mv^2/r . Slipping begins when $mv^2/r = \mu_s mg$, i.e. $a_c = \mu_s g$.
- Cones and banked tracks (smooth):** the vertical component of N balances mg , and the horizontal component provides mv^2/r .

EXAMPLE 6.1 **SJPO 2018 Q7**

7. A bug is just about to slip on a circular turntable of radius R rotating at constant angular velocity. The bug is halfway between the centre and the edge and the coefficient of static friction is $1/4$. What is the acceleration of the bug?
- A. $3g$
B. $2g$
C. $3g/2$
D. $g/2$
E. $g/4$

WORKING**Answer** _____**EXAMPLE 6.2** **SJPO 2024 Q1**

1. A smooth hollow cone of half-angle θ is inverted with its axis vertical. A particle of mass m rolls around in a horizontal circle within the inner surface of the cone. What is the magnitude of the normal force N exerted by the cone on the particle?



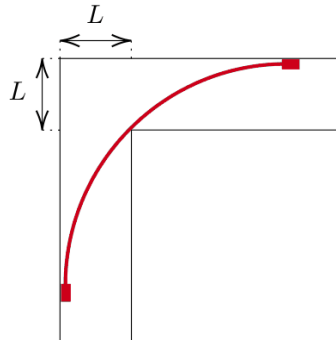
- A. $mg \cos \theta$
B. $mg \sin \theta$
C. $\frac{mg}{\cos \theta}$
D. $\frac{mg}{\sin \theta}$
E. $mg \tan \theta$

WORKING**Answer** _____

Racing line. A larger radius allows a larger speed for the same friction: $v_{\max} = \sqrt{\mu g r}$. On a bend of finite width, the best path is the **largest circle that fits** inside the road.

EXAMPLE 6.3 SJPO 2024 Q18

18. A car drives at a constant speed along a horizontal road of width L that has a sharp 90° bend as shown in the figure. The coefficient of static friction between the car and the road is μ . What is the maximum speed at which the car can negotiate the bend without slipping? Neglect the size of the car.



- A. $\sqrt{\mu g L}$
- B. $\sqrt{2\mu g L}$
- C. $\sqrt{\sqrt{2}\mu g L}$
- D. $\sqrt{(2 + \sqrt{2})\mu g L}$
- E. $\sqrt{(2 + 2\sqrt{2})\mu g L}$

WORKING

Answer _____

6.2 Vertical and Non-Uniform Circular Motion

In a vertical circle the speed changes, so use **two** equations:

1. **Energy conservation** gives v at any position.
2. **The radial equation:** (net force towards the centre) = $\frac{mv^2}{r}$ gives the tension or normal force.

PENDULUM AT ANGLE θ FROM THE VERTICAL

$$T - mg \cos \theta = \frac{mv^2}{L}$$

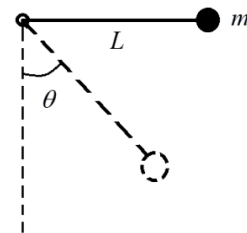
Tension is greatest at the lowest point, where the speed is greatest and gravity points straight away from the centre.

Sudden change of radius (a string catching on a peg): the speed is unchanged at that instant, but mv^2/r jumps because r changed.

EXAMPLE 6.4 SJPO 2022 Q10–11

Questions 10 and 11.

10. A simple pendulum, consisting of a string tied to a frictionless fixed pivot at one end and a bob of mass m at the other end, is gently released from the horizontal position (i.e. $\theta = 90^\circ$) as shown.



After the release, the tension of the string is given by ____.

- a) $3mg \cos \theta$
- b) $3mg \sin \theta$
- c) $mg \cos \theta$
- d) $mg \sin \theta$
- e) $4mg \cos \theta$

11. In order for the string not to break during the swing, it must be able to withstand a tension of ____.

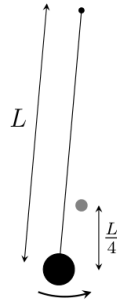
- a) $2mg$
- b) $4mg$
- c) $8mg$
- d) $6mg$
- e) $3mg$

WORKING

Answer _____

Question 14

A pendulum of mass m and length L swings in the vertical plane. At the lowest point of its motion, the string catches on a peg located at a distance $L/4$ above the mass. If the tension in the string just before hitting the peg is $T = 3mg$, what is the tension in the string right after hitting the peg? Neglect air resistance.



- (a) $\frac{3}{4}mg$
- (b) $\frac{3}{2}mg$
- (c) $3mg$
- (d) $\frac{5}{2}mg$
- (e) $9mg$

WORKING

Answer _____

Non-uniform circular motion

If the speed changes, the acceleration has two perpendicular parts: tangential $a_t = r\alpha$ and centripetal $a_c = r\omega^2$, so $a = \sqrt{a_t^2 + a_c^2}$. For a turntable spun up from rest with constant α , $\omega = \alpha t$.

7 Gravitation

Newton's law of gravitation, circular orbits, and the energy bookkeeping that explains why a satellite speeds up as drag pulls it down.

7.1 Gravitational Force and Circular Orbits

7.2 Gravitational Potential Energy

LEARNING OUTCOMES

Candidates should be able to:

- apply Newton's law of gravitation $F = GMm/r^2$ and the relation $g = GM/R^2$
- derive orbital speed and period for circular orbits, including geostationary orbits and Kepler's third law
- use gravitational potential energy $U = -GMm/r$ for large changes in separation
- show that for a circular orbit $E = -K = \frac{1}{2}U$, and explain how drag changes the orbit
- locate the point between two bodies where their gravitational fields cancel

7.1 Gravitational Force and Circular Orbits

NEWTON'S LAW OF GRAVITATION

$$F = \frac{GMm}{r^2} \quad g = \frac{GM}{r^2} \text{ (field strength)} \quad g_{\text{surface}} = \frac{GM}{R^2} \Rightarrow GM = gR^2$$

The swap $GM = gR^2$ is very useful when G or M is not given.

Circular orbits

Gravity supplies the centripetal force:

$$\frac{GMm}{r^2} = \frac{mv^2}{r} = mr\omega^2 \Rightarrow v = \sqrt{\frac{GM}{r}}, \quad T^2 = \frac{4\pi^2}{GM}r^3$$

The second result is **Kepler's third law**. It also holds for ellipses if r is replaced by the semi-major axis a . A **geostationary** orbit has $T = 24$ h, lies above the equator, and moves in the same direction as Earth's rotation.

EXAMPLE 7.1 **SJPO 2018 Q8**

8. The height of a circular geosynchronous orbit (24 hr period) from the earth's surface is approximately ____ Earth radii. (Hint: You may need the gravitational constant, mass of the Earth and the gravitational acceleration at the surface of the Earth. These are given in the list of constants.)
- A. 3
B. 4
C. 5
D. 6
E. 7

WORKING**Answer** _____**Energy of a circular orbit**

$$K = \frac{GMm}{2r}, \quad U = -\frac{GMm}{r}, \quad E = K + U = -\frac{GMm}{2r} = -K$$

Because $E = -K$, when drag *removes* energy, E becomes more negative, r decreases and K **increases**. The satellite spirals inward and speeds up. This is the "satellite drag paradox".

Question 24

A satellite of mass m is in a circular orbit of radius r around a planet of mass M . It can be shown that its kinetic energy and potential energy are given by

$$\begin{aligned}\text{Kinetic energy} &= \frac{GMm}{2r} \\ \text{Potential energy} &= -\frac{GMm}{r}\end{aligned}$$

Suppose that the satellite's engine is suddenly shut down. As a result of resistive forces, how do the radius of the orbit and the linear speed of the satellite change over time?

- (a) Radius decreases and speed decreases.
- (b) Radius decreases and speed increases.
- (c) Radius increases and speed decreases.
- (d) Radius increases and speed increases.
- (e) Radius remains unchanged and speed decreases.

WORKING

Answer _____

7.2 Gravitational Potential Energy

For large changes in separation, $mg\Delta h$ is not accurate. Use

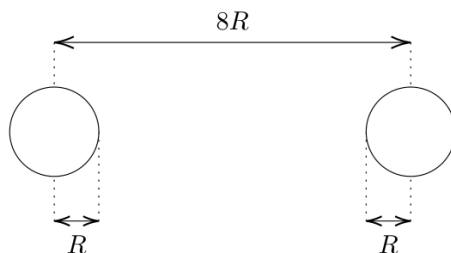
GRAVITATIONAL POTENTIAL ENERGY (ZERO AT INFINITY)

$$U = -\frac{GMm}{r} \quad \Delta U = GMm \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \text{ when moving from } r_1 \text{ to } r_2$$

- Two bodies released from rest move towards each other. Conserve both momentum (equal masses have equal speeds) and energy.
- Two spheres "collide" when their centres are $R_1 + R_2$ apart, not when $r = 0$.

EXAMPLE 7.3 **SJPO 2024 Q23**

23. In an experiment gone *very* wrong, Shuhao accidentally created a clone of Earth with the same mass and radius R as our Earth. The two Earths are initially stationary with a separation of $8R$ between their centres. As they attract each other due to gravity, they eventually crash into each other. What is the velocity of each Earth at impact?



- A. $\sqrt{\frac{1}{6}gR}$
B. $\sqrt{\frac{3}{8}gR}$
C. $\sqrt{\frac{3}{4}gR}$
D. $\sqrt{6gR}$
E. $\sqrt{12gR}$

WORKING**Answer** _____**Null point between two masses**

Between masses M_1 and M_2 a distance d apart, the fields cancel at distance x from M_1 where

$$\frac{GM_1}{x^2} = \frac{GM_2}{(d-x)^2} \Rightarrow x = \frac{d}{1 + \sqrt{M_2/M_1}}$$

A mass released just beyond this point falls towards the farther body instead.

Question 23

A mass is initially held at a height h above the Earth's surface, with the Sun directly overhead. The mass is then released. What is the minimum height h such that the mass proceeds to rise towards the Sun instead of falling back towards the Earth?

Data:

- Mass of Sun $M_S = 1.99 \times 10^{30}$ kg
- Mass of Earth $M_E = 5.97 \times 10^{24}$ kg
- Radius of Sun $R_S = 6.96 \times 10^5$ km
- Radius of Earth $R_E = 6.37 \times 10^3$ km
- Distance between the centres of the Sun and Earth $d_{SE} = 1.50 \times 10^8$ km

- (a) 4.43×10^2 km
- (b) 2.59×10^4 km
- (c) 2.53×10^5 km
- (d) 1.49×10^6 km
- (e) 7.50×10^7 km

WORKING

Answer _____

8 Rotational Dynamics

Rigid bodies that spin as well as move: moment of inertia, torque, rolling without slipping, and angular momentum.

8.1 Moment of Inertia and Torque

8.3 Angular Momentum and Angular Impulse

8.2 Rolling Without Slipping

LEARNING OUTCOMES

Candidates should be able to:

- use moments of inertia of standard bodies and reason about I by symmetry
- apply $\tau = I\alpha$ together with $F = ma$, including pulleys with mass
- apply the rolling condition $v = R\omega$ and split kinetic energy into translational and rotational parts
- identify which forces do work on a rolling body
- relate angular impulse to change in angular momentum, and find the direction of $\vec{\omega}$ and $\vec{L}$

8.1 Moment of Inertia and Torque

Rotational quantities mirror linear ones:

Linear	Rotational
x, v, a	θ, ω, α (with $v = r\omega, a_t = r\alpha$)
mass m	moment of inertia $I = \sum m_i r_i^2$
$F = ma$	$\tau = I\alpha$
$K = \frac{1}{2}mv^2$	$K = \frac{1}{2}I\omega^2$
$p = mv$	$L = I\omega$

STANDARD MOMENTS OF INERTIA (ABOUT THE AXIS THROUGH THE CM)

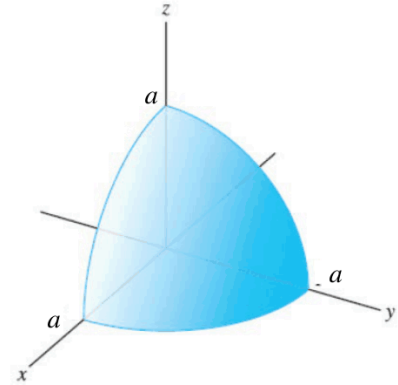
solid sphere $\frac{2}{5}MR^2$ solid disc / cylinder $\frac{1}{2}MR^2$ ring / hoop MR^2 rod (centre) $\frac{1}{12}ML^2$

Symmetry argument. I depends only on how far each mass element is from the axis. Cut a sphere into 8 identical octants: each octant has its mass at the same distances from an axis through the centre as the whole sphere does. So an octant of mass m has $I = \frac{2}{5}ma^2$ about the same axis.

EXAMPLE 8.1 **SJPO 2018 Q9**

9. The moment of inertia of a solid sphere of mass M and radius a , about an axis passing through its centre, is $(2/5)Ma^2$. The mass of a solid uniform octant (one-eighth) of a sphere of radius a , is m . The moment of inertia about an axis along one of the straight edges (e.g. z axis) is ____.

- A. $\frac{4}{3}ma^2$
B. $\frac{2}{5}ma^2$
C. $\frac{3}{8}ma^2$
D. $\frac{1}{8}ma^2$
E. $\frac{1}{20}ma^2$

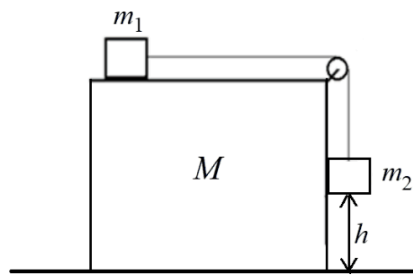
**WORKING****Answer** _____**Pulleys with mass**

If the pulley has moment of inertia I , the tensions on either side are **different**: $(T_2 - T_1)R = I\alpha = I\frac{a}{R}$. For a uniform disc pulley this gives $T_2 - T_1 = \frac{1}{2}m_p a$. The pulley behaves like an extra mass $\frac{1}{2}m_p$ that has to be accelerated.

EXAMPLE 8.2 SJPO 2022 Q21

For questions 17- 21

A block of mass m_1 is placed on top of a block of mass M . Holding block m_1 stationary, it is joined by a string which passes over the pulley to a block of mass m_2 hanging by the side of (and in contact with) block M , and is at height h above the floor. We will assume that the string is inextensible and massless, and all the contacts between objects in relative motion are frictionless. Furthermore, we will be concerned with the part of the motion before m_1 reaches the pulley.



21. For this question, assume that M is fixed to the floor, and the mass and radius of the pulley are m_3 and R , respectively. What is the acceleration of blocks m_1 and m_2 ? Hint: Take the moment of inertia of the pulley as $I = \frac{1}{2}m_3R^2$

- a) $\frac{m_1}{m_1+m_2+\frac{m_3}{2}}g$
- b) $\frac{m_2}{m_1+m_2+\frac{m_3}{2}}g$
- c) $\left(1 - \frac{m_3}{2(m_1+m_2)}\right)g$
- d) $\frac{m_1+m_2+\frac{m_3}{2}}{m_1}g$
- e) $\frac{m_1+m_2+\frac{m_3}{2}}{m_2}g$

WORKING

Answer _____

8.2 Rolling Without Slipping

ROLLING WITHOUT SLIPPING

$$v_{\text{cm}} = R\omega \quad a_{\text{cm}} = R\alpha \quad K = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$$

- The contact point is instantaneously at rest. The top of the wheel moves at $2v$.

- **Static friction** at the contact point does **no work** on a body rolling without slipping, because the contact point does not move. Yet it is the force that provides the torque to change ω .
- For $I = kMR^2$: $K_{\text{rot}}/K_{\text{trans}} = k$. For a solid sphere ($k = \frac{2}{5}$), rotation holds $\frac{2}{7}$ of the total KE.

Work by a force applied at a point on the body: $W = F \times (\text{distance moved by the point of application})$.

A string unwinding from the top of a rolling cylinder moves its end $2x$ while the centre moves x .

EXAMPLE 8.3 SJPO 2018 Q4

4. A sphere rolls without slipping down a rough inclined plane. The gain in rotational kinetic energy is due **directly** to the work done by _____.
- Static friction
 - Kinetic friction
 - Weight
 - Normal contact force
 - Air resistance

WORKING

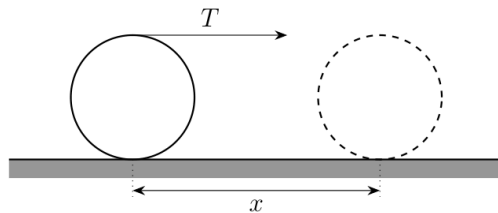
Answer _____

EXAMPLE 8.4 SJPO 2026 Q17–18

Use the information below to answer Questions 17 and 18.

A uniform solid cylinder lies on rough horizontal ground. A string wrapped around the cylinder is pulled at the top with constant horizontal tension T . The cylinder rolls without slipping, moving a distance x forward.

(The moment of inertia of a uniform solid cylinder of mass m and radius r is $I = \frac{1}{2}mr^2$ about its central axis.)

**Question 17**

What is the magnitude of the friction exerted on the cylinder by the ground?

- (a) 0
- (b) $\frac{1}{4}T$
- (c) $\frac{1}{3}T$
- (d) $\frac{1}{2}T$
- (e) T

Question 18

What is the work done by the tension T ?

- (a) 0
- (b) $\frac{1}{2}Tx$
- (c) Tx
- (d) $\frac{3}{2}Tx$
- (e) $2Tx$

WORKING

Answer _____

8.3 Angular Momentum and Angular Impulse

ANGULAR MOMENTUM

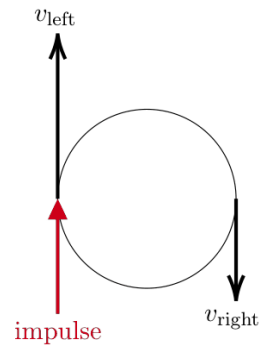
$$\vec{L} = \vec{r} \times \vec{p} \quad L = I\omega \text{ (rigid body about a fixed or CM axis)} \quad \vec{\tau} = \frac{d\vec{L}}{dt}$$

- A brief impulse J applied at perpendicular distance d from the CM changes linear momentum by J **and** angular momentum about the CM by Jd .
- Velocity of a point on a rigid body = velocity of the CM + velocity due to rotation about the CM (ωr , tangential).
- Directions: $\vec{\omega}$ lies along the axis of rotation (right-hand rule). $\vec{L} = \sum \vec{r} \times \vec{p}$ need **not** be parallel to $\vec{\omega}$ when the body is not symmetric about the rotation axis, for example a tilted stick swinging round a vertical axis.

EXAMPLE 8.5 **SJPO 2024 Q20**

20. A uniform solid disc rests flat on a frictionless table. A brief tangential impulse is applied on the leftmost point of the disc. What is the resulting ratio of speeds of the leftmost and rightmost points of the disc, $\frac{|v_{\text{left}}|}{|v_{\text{right}}|}$?

(The moment of inertia of a uniform solid disc of mass M and radius R about its central axis is given by $I = \frac{1}{2}MR^2$.)

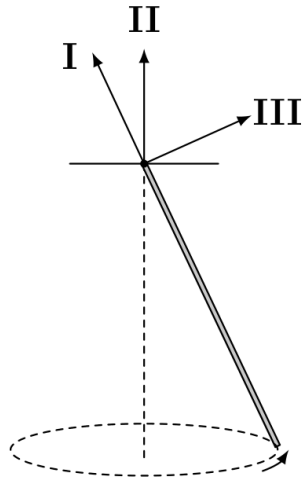


- A. 1
- B. $\frac{5}{3}$
- C. 2
- D. 3
- E. 4

WORKING**Answer** _____

Question 19

A thin stick is pivoted at its top end and is set to swing rigidly in uniform circular motion about the vertical axis. What are the directions of its angular velocity $\vec{\omega}$ and its angular momentum $\vec{L}$?



- (a) $\vec{\omega}$ in direction **I** and $\vec{L}$ in direction **I**
- (b) $\vec{\omega}$ in direction **II** and $\vec{L}$ in direction **I**
- (c) $\vec{\omega}$ in direction **II** and $\vec{L}$ in direction **II**
- (d) $\vec{\omega}$ in direction **II** and $\vec{L}$ in direction **III**
- (e) $\vec{\omega}$ in direction **III** and $\vec{L}$ in direction **II**

WORKING**Answer** _____

9 Fluids

Pressure, buoyancy and flow: Archimedes' principle for floating and sinking, continuity and Bernoulli's equation for moving fluids.

9.1 Pressure and Buoyancy

9.2 Fluid Flow: Continuity and Bernoulli

LEARNING OUTCOMES

Candidates should be able to:

- use $p = p_0 + \rho gh$ for pressure in a static fluid
- apply Archimedes' principle to floating and submerged bodies
- reason about changes of water level when floating ice melts
- apply the continuity equation $A_1 v_1 = A_2 v_2$
- apply Bernoulli's equation, including stagnation pressure and Pitot-tube arguments

9.1 Pressure and Buoyancy

HYDROSTATICS

$$p = p_0 + \rho gh \quad \text{Upthrust } U = \rho_{\text{fluid}} V_{\text{submerged}} g$$

Archimedes' principle. The upthrust on a body equals the weight of fluid it displaces.

- Floating** body: upthrust = weight, so $\frac{V_{\text{sub}}}{V} = \frac{\rho_{\text{body}}}{\rho_{\text{fluid}}}$.
- Maximum load before sinking: when the whole volume is submerged, $W_{\text{max}} = \rho_{\text{fluid}} V g$ (minus the body's own weight).
- Specific gravity** = density relative to pure water (1000 kg m^{-3}).

EXAMPLE 9.1 **SJPO 2022 Q24**

24. How much weight can a Styrofoam cup of volume 200 cm^3 carry before it sinks in water? Assume the weight of Styrofoam is negligible and take the density of water as $1 \text{ g}\cdot\text{cm}^{-3}$.

- a) 0.2N
- b) 5N
- c) 20N
- d) 1N
- e) 2N

WORKING

Answer _____

Melting ice

Floating ice displaces its own **mass** of the liquid it floats in. When it melts, it becomes the same mass of meltwater. Compare the volume of that meltwater with the volume originally displaced:

- Pure ice in pure water: same density, so the level is unchanged.
- Pure ice in denser seawater: the meltwater (density 1.00) takes up *more* volume than the seawater displaced (1.03), so the level **rises**.

EXAMPLE 9.2 SJPO 2022 Q25–26

For questions 25 and 26

The specific gravity of a substance is defined as the ratio of the density of the substance to that of pure water. The specific gravity of pure ice is 0.92, and that of seawater is 1.03.

25. What percentage of an iceberg made up of pure ice is above the surface of the seawater?

- a) 9%
- b) 8%
- c) 12%
- d) 11%
- e) 10%

26. A piece of pure ice is gently placed into a measuring cylinder containing seawater. The ice floats and the seawater level is noted. After the ice has melted, the seawater level will ____.

- a) rise
- b) stay the same
- c) drop
- d) depend on the amount of the pure ice, i.e. can be either a) or b) or c)

WORKING

Answer _____

9.2 Fluid Flow: Continuity and Bernoulli

For steady flow of an incompressible fluid:

CONTINUITY AND BERNOULLI

$$A_1 v_1 = A_2 v_2 \quad p + \frac{1}{2} \rho v^2 + \rho g h = \text{constant along a streamline}$$

- **Continuity:** where the flow is faster, the stream is narrower. Water speeding up as it falls from a tap gets thinner.
- **Stagnation point:** where moving fluid is brought to rest (the open end of a Pitot / L-shaped tube facing the flow), the pressure rises by $\frac{1}{2} \rho v^2$. The water in the tube stands $h = \frac{v^2}{2g}$ above the free

surface.

- **Large tank:** the speed of the free surface is negligible. Efflux speed from depth l is $\sqrt{2gl}$ (Torricelli).
- Pressure in a liquid can't drop below zero (the liquid would boil or break). That limit sets, for example, the maximum height of a siphon.

EXAMPLE 9.3 SJPO 2022 Q22

22. Water is flowing vertically downward from a tap at a moderate speed. As it falls, the water column's diameter ____.

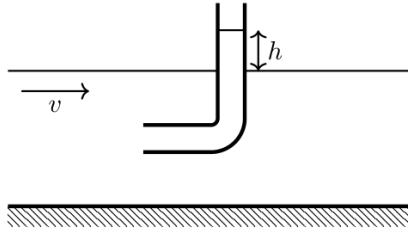
- a) decreases
- b) remains unchanged
- c) increases
- d) decreases in Northern Hemisphere but increases in Southern Hemisphere
- e) increases in Northern Hemisphere but decreases in Southern Hemisphere

WORKING

Answer _____

EXAMPLE 9.4 **SJPO 2024 Q24**

24. An L-shaped pipe is placed into a river flowing at speed v , with the open end facing the current head-on. The water in the pipe is observed to rise to a height $h = 0.50$ m above the water surface of the river. Find v .



- A. 3.1 m s^{-1}
- B. 4.2 m s^{-1}
- C. 5.6 m s^{-1}
- D. 6.2 m s^{-1}
- E. 7.0 m s^{-1}

WORKING**Answer** _____

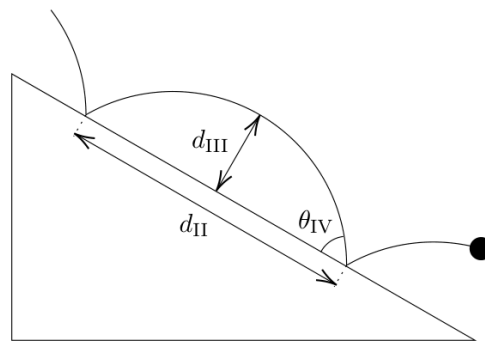
Tutorial Questions

15 past-year SJPO questions covering the whole topic, roughly in chapter order. Attempt them under timed conditions (about 2.5 minutes each, as in the real paper). Full worked solutions are in the Solutions booklet.

QUESTION T1 SJPO 2024 Q10

Ch 1 · Kinematics

10. A ball is dropped from rest onto an inclined slope. Assuming there are no energy losses, which of the following statements are true about the ball's motion as it bounces down the ramp?
- I. The time interval between each bounce decreases over time.
 - II. The distance between each bounce, d_{II} , increases over time.
 - III. The maximum distance reached by the ball from the ramp between each bounce, d_{III} , remains constant over time.
 - IV. The angle of the ball's trajectory from the ramp at each bounce, θ_{IV} , remains constant over time.



- A. I and II only
- B. II and III only
- C. III and IV only
- D. I, II and III only
- E. I, III and IV only

QUESTION T2 SJPO 2026 Q8

Ch 1 · Kinematics

Question 8

A droplet falls through a tree and collides with a leaf every vertical distance h . Each collision causes the droplet to lose half of its speed. After falling through a large distance, the droplet achieves steady state. What is the terminal speed reached by the droplet? Neglect air resistance.

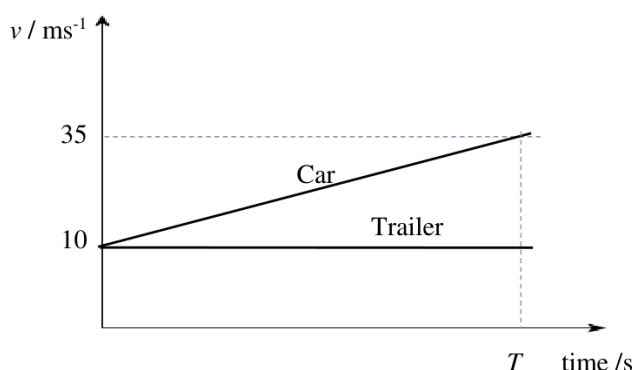
- (a) $\sqrt{gh}$
- (b) $\sqrt{2gh}$
- (c) $\sqrt{\frac{2}{3}gh}$
- (d) $\sqrt{\frac{8}{3}gh}$
- (e) $\sqrt{8gh}$

11. The minimum time T for a car to safely overtake a long trailer is measured from the time the front of the car is level with the rear of the trailer, until the rear of the car is one full car-length ahead of the trailer.

The car is 3.5 m long and the trailer is 15.0 m long.

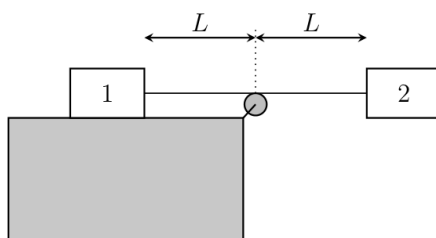
The graph shows the speed-time graphs of the car and the trailer. What is the minimum time T ?

- A. 2.16 s
- B. 1.76 s
- C. 1.48 s
- D. 0.88 s
- E. 0.64 s



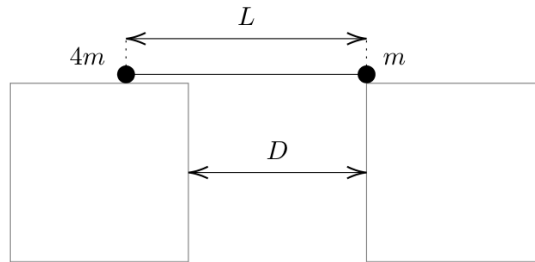
Question 12

Two blocks of equal masses are connected by an inextensible string passing over a light pulley at the edge of a table. Block 1 is on the table, while block 2 is suspended freely. Both blocks are initially held at equal horizontal distances from the pulley. The system is then released from rest. Which statement is true about the motion of the blocks? Neglect friction and air resistance.



- (a) Both blocks remain at rest as their masses are equal.
- (b) Block 1 reaches the pulley before block 2 strikes the table.
- (c) Block 1 reaches the pulley after block 2 strikes the table.
- (d) Block 1 reaches the pulley and block 2 strikes the table at the same time.
- (e) Not enough information to determine which event occurs first.

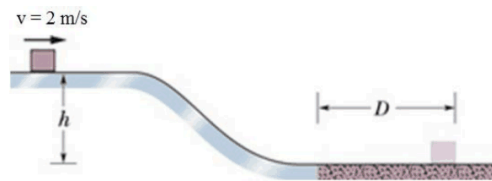
16. Brian wants to bridge two cliffs using a plank of length L . The plank itself is essentially massless, but there is a mass m attached to its front and a mass $4m$ attached to its back. The bridge is formed by pushing the plank horizontally across the gap, until the full gap is covered by the plank. What is the maximum gap D between the cliffs that can be bridged in this manner?



- A. $\frac{1}{5}L$
- B. $\frac{1}{2}L$
- C. $\frac{3}{5}L$
- D. $\frac{3}{4}L$
- E. $\frac{4}{5}L$

For questions 14 – 16

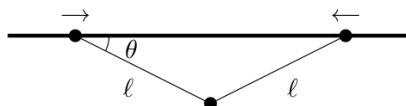
A block of mass m slides with an initial velocity of 2 m/s along a horizontal track that descends through distance h . The track is frictionless except for the lower section. There the block slides to a stop in a certain distance D because of friction.



14. If h is halved, D ____.
- will be halved
 - will be unchanged
 - will be doubled
 - will be tripled
 - cannot be determined due to insufficient information
15. If m is doubled, D ____.
- will be halved
 - will be unchanged
 - will be doubled
 - will be tripled
 - cannot be determined due to insufficient information
16. If m is doubled, the total thermal energy generated due to friction as the block slides to a stop ____.
- is halved
 - is unchanged
 - is doubled
 - is tripled
 - cannot be determined due to insufficient information

QUESTION T7 SJPO 2024 Q14

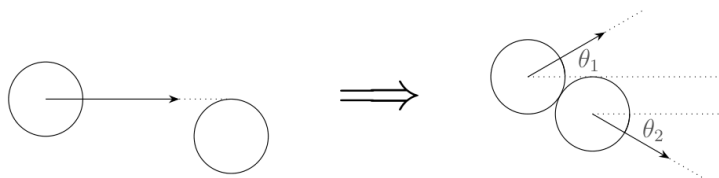
14. Three identical point masses m are connected by strings of length ℓ in the vertical plane, as shown below. The two top masses are constrained to move horizontally along a frictionless track. If the system is released from rest at an initial angle $\theta = 0$ from the horizontal, find the relative velocity of the two top masses just before they collide with each other.



- 0
- $\sqrt{g\ell}$
- $\sqrt{2g\ell}$
- $\sqrt{3g\ell}$
- $\sqrt{4g\ell}$

Question 16

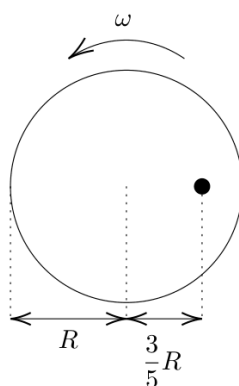
Two identical billiard balls undergo a side-on elastic collision on a frictionless table, where one ball strikes another stationary ball off-centre. The moving ball is deflected by an angle $\theta_1 = 75^\circ$. What is the deflection angle θ_2 of the other ball?



- (a) 15°
- (b) 30°
- (c) 45°
- (d) 60°
- (e) 75°

QUESTION T9 SJPO 2024 Q17

17. A horizontal turntable of radius R rotates with constant angular velocity ω . An ant, positioned a distance $\frac{3R}{5}$ from the centre, initially rotates along with the turntable. All of a sudden, it decides to jump vertically (relative to itself). What is the minimum duration of its jump such that it escapes the turntable?



- A. $\frac{2}{5\omega}$
- B. $\frac{3}{5\omega}$
- C. $\frac{4}{5\omega}$
- D. $\frac{4}{3\omega}$
- E. $\frac{5}{3\omega}$

Question 20

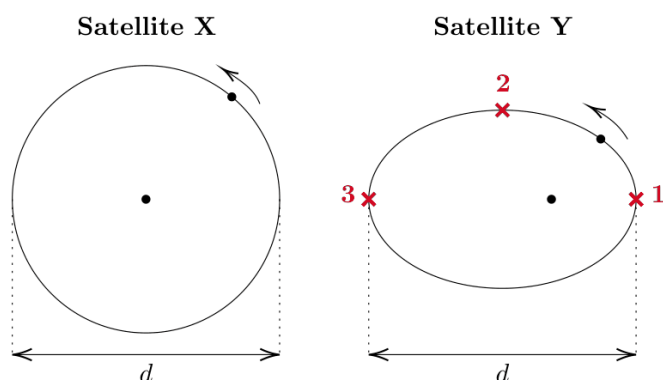
A mass is placed on the edge of a horizontal turntable of radius R and coefficient of static friction μ_s . Starting from rest (at time $t = 0$), the turntable begins to rotate at a constant angular acceleration α . At what time t does the mass slip off the turntable?

- (a) $\frac{\mu_s g}{R\alpha}$
- (b) $\left(\frac{\mu_s g}{R\alpha^2}\right)^{1/2}$
- (c) $\left(\frac{\mu_s g}{R\alpha^2} - \frac{1}{\alpha}\right)^{1/2}$
- (d) $\left(\frac{\mu_s^2 g^2}{R^2 \alpha^2} - 1\right)^{1/2}$
- (e) $\left(\frac{\mu_s^2 g^2}{R^2 \alpha^4} - \frac{1}{\alpha^2}\right)^{1/4}$

QUESTION T11 SJPO 2024 Q22

22. Two satellites X and Y orbit the same planet and have equal masses. The orbit of satellite X is circular, whereas the orbit of satellite Y is elliptical. The major axis of satellite Y's orbit has the same length as the diameter d of satellite X's orbit. Which of the following statements are true?

- I. The period of the orbit is equal for both satellites.
- II. The total energy of the orbit is equal for both satellites.
- III. The angular momentum of the orbit (with respect to the planet) is equal for both satellites.
- IV. The speeds of Satellite Y at points 1, 2 and 3 of its orbit (as shown below) are related by $v_1 < v_2 < v_3$.



- A. III only
- B. I and II only
- C. I and IV only
- D. III and IV only
- E. II, III and IV only

Question 22

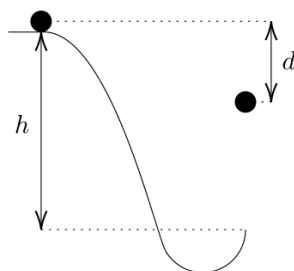
A tiny dust particle is at rest in space at a large distance r from the Earth. Due to Earth's gravitational attraction, the dust particle begins moving in a straight line towards the Earth. The Earth has mass M and radius R , with $r \gg R$. What is the time taken for the dust particle to reach the Earth?

- (a) $\sqrt{\frac{2r}{g}}$
- (b) $\sqrt{\frac{2r^3}{GM}}$
- (c) $\frac{\pi}{2\sqrt{2}}\sqrt{\frac{r^3}{GM}}$
- (d) $\pi\sqrt{\frac{r^3}{GM}}$
- (e) $2\pi\sqrt{\frac{r^3}{GM}}$

QUESTION T13 SJPO 2024 Q21

21. A uniform solid sphere is placed at the top of a valley and released from rest. It rolls down, and upon reaching the end of the valley, a height h below the top of the valley, it rises upward into the air. Throughout its motion, the sphere never experiences any energy dissipation and never slips. Yet, Chris claims that the sphere will never return to its original height, falling short by a distance d . Is Chris right, and if so, what is d ?

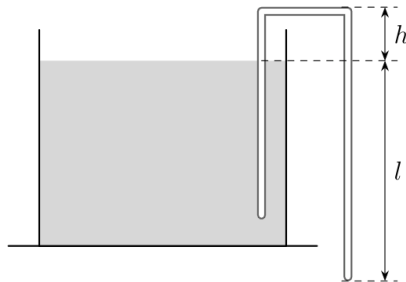
(The moment of inertia of a uniform solid sphere of mass M and radius R about its central axis is given by $I = \frac{2}{5}MR^2$.)



- A. Chris is wrong, as the sphere should return to its original height.
- B. Chris is right, but there is not enough information to calculate d .
- C. Chris is right, $d = \frac{1}{10}h$.
- D. Chris is right, $d = \frac{2}{7}h$.
- E. Chris is right, $d = \frac{2}{5}h$.

Question 21

A siphon is a seemingly gravity-defying device that can empty a tank of fluid by causing fluid to flow up, then down, and out of the siphon. In this setup, the siphon reaches a maximum height h above the fluid surface, and minimum height l below the fluid surface. Let P_0 be the atmospheric pressure and ρ be the fluid density. What is the maximum height h for which the siphon will work? Assume that the siphon has a uniform diameter and that the tank is very large.



- (a) $\frac{P_0}{\rho g}$
- (b) $\frac{P_0}{2\rho g}$
- (c) $\frac{P_0}{\rho g} - l$
- (d) $\frac{P_0}{2\rho g} - l$
- (e) There is no maximum height; the siphon will always work for any h .

Question 5

A bubble of uniform density ρ_b and volume V is deep within a fluid of uniform density ρ_f . What is the total change in potential energy of the system if the bubble rises by a small height h ?

- (a) $\rho_b V g h$
- (b) $\rho_f V g h$
- (c) $(\rho_b - \rho_f) V g h$
- (d) $(\rho_f - \rho_b) V g h$
- (e) $(\rho_b + \rho_f) V g h$